WORKED SOLUTION

STEP III 2014, Question 3

Pure • Distance to a Parabola

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/14-s3-q3

The key idea

Every point of the parabola can be written as $(at^2, 2at)$, so each distance we need becomes a function of the single variable t, and in both parts that function turns out to be a quadratic (in t for the line, in t^2 for the point). Minimising a quadratic is just completing the square, but we must respect where the variable is allowed to live: t^2 can't be negative, and that restriction is exactly what splits part (ii) into cases. The circle then comes for free, because the distance from a point to a circle is its distance to the centre minus the radius.

Solution

Part (i)

Write L as $mx - y + c = 0$. The perpendicular distance from the point $P = (at^2, 2at)$ of the parabola to L is

$$\frac{|mat^2 - 2at + c|}{\sqrt{m^2+1}}.$$

Let $f(t) = mat^2 - 2at + c$. Since $ma > 0$, completing the square gives

$$f(t) = ma\left(t - \frac{1}{m}\right)^2 + c - \frac{a}{m} = ma\left(t - \frac{1}{m}\right)^2 + \frac{mc - a}{m},$$

so f has minimum value $\dfrac{mc-a}{m}$, attained at $t = 1/m$.

Case $mc > a$**.** Then $f(t) \geq \frac{mc-a}{m} > 0$ for every t, so the whole parabola lies on one side of L and never meets it. The shortest distance between L and the parabola is the least distance from a point of the parabola to L, which is the least value of $|f(t)|/\sqrt{m^2+1} = f(t)/\sqrt{m^2+1}$:

$$\frac{mc-a}{m\sqrt{m^2+1}},$$

as required. The nearest point is $t = 1/m$, where the tangent to the parabola has gradient $\mathrm{d}y/\mathrm{d}x = 1/t = m$: it is parallel to L, just as geometry suggests.

Case $mc \leq a$**.** Now $f(1/m) \leq 0$, while $f(t) \to +\infty$ as $t \to \infty$. Since f is continuous, $f(t) = 0$ for some t, so L meets the parabola and the shortest distance is 0.

Part (ii): the point $(p, 0)$

The square of the distance from $C = (p, 0)$ to $P = (at^2, 2at)$ is

$$(at^2 - p)^2 + 4a^2t^2.$$

This depends only on $u = t^2$, and u takes every value in $[0, \infty)$. So we minimise

$$g(u) = (au - p)^2 + 4a^2u = a^2u^2 - 2a(p - 2a)u + p^2 \qquad (u \geq 0).$$

Completing the square,

$$g(u) = a^2\left(u - \frac{p - 2a}{a}\right)^2 + p^2 - (p - 2a)^2 = a^2\left(u - \frac{p - 2a}{a}\right)^2 + 4a(p - a).$$

Ask yourself The unrestricted minimum of g is at $u = (p - 2a)/a$. Is that value always available?

No: $u = t^2$ must be non-negative, and $(p - 2a)/a \geq 0$ only when $p \geq 2a$.

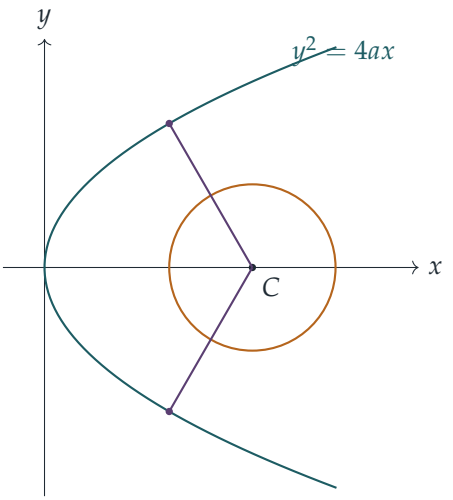
Case $p/a > 2$**.** The vertex $u = p/a - 2$ is positive, so it is allowed, and the least value of g is $4a(p - a)$, which is positive since $p > 2a > a$. The shortest distance is

$$2\sqrt{a(p - a)},$$

achieved at the two points $(p - 2a,\ \pm 2\sqrt{a(p - 2a)})$.

Case $p/a \leq 2$**.** The vertex is at $u \leq 0$, so g is increasing on $u \geq 0$ and its least value there is $g(0) = p^2$. The shortest distance is p, achieved at the vertex $(0, 0)$ of the parabola.

The two answers agree at $p = 2a$, where both give $2a$.



The picture shows $a = 1$, $p = 5$: the nearest points of the parabola to C (purple) are at distance $2\sqrt{1 \cdot 4} = 4$, and a circle of radius $b = 2$ about C (amber) stays a distance 2 away.

Part (ii): the circle

Let d be the distance just found from $C = (p, 0)$ to the parabola, so

$$d = \begin{cases} p & \text{if } p \leq 2a, \\ 2\sqrt{a(p-a)} & \text{if } p > 2a. \end{cases}$$

The circle has centre C and radius b.

If $d > b$. Take any point P on the parabola and Q on the circle. By the triangle inequality,

$$|PQ| \geq |PC| - |CQ| = |PC| - b \geq d - b > 0.$$

Equality holds when P is a nearest point of the parabola to C and Q is where the segment CP crosses the circle (it does cross, as $|CP| = d > b$). So the shortest distance is $d - b$.

If $d \leq b$. As P moves along the parabola, $|PC|$ varies continuously from its least value $d \leq b$ to arbitrarily large values, so $|PC| = b$ somewhere. That point lies on both curves, and the shortest distance is 0.

Putting this together, the shortest distance between the circle and the parabola is

$$\begin{cases} p - b & \text{if } p \leq 2a \text{ and } b < p, \\ 2\sqrt{a(p-a)} - b & \text{if } p > 2a \text{ and } b < 2\sqrt{a(p-a)}, \\ 0 & \text{otherwise (that is, if } b \geq d). \end{cases}$$

Checks

- Take $m = 1$, $c = 2$, $a = 1$, so $mc = 2 > a$. The formula gives $\frac{1}{\sqrt{2}} \approx 0.7071$; directly, the point $t = 1$ is $(1, 2)$, and its distance to $x - y + 2 = 0$ is $|1 - 2 + 2|/\sqrt{2} = 1/\sqrt{2}$. If instead $c = \frac{1}{2}$ (so $mc < a$), the line $y = x + \frac{1}{2}$ meets $y^2 = 4x$ where $x^2 - 3x + \frac{1}{4} = 0$, which has real roots, confirming distance 0.
- With $a = 1$, $p = 5$: the points $(3, \pm 2\sqrt{3})$ are at distance $\sqrt{4 + 12} = 4 = 2\sqrt{1 \cdot 4}$ from $(5, 0)$, and a computer search over the parabola found nothing closer.
- As $p \to 0$ the distance $p \to 0$, correctly, since $(0, 0)$ is on the parabola. For large p, $2\sqrt{a(p - a)} < p$, as it must be, because the vertex is at distance p.
- All the results were confirmed by a computer search over hundreds of random choices of m, c, a, p and b.

Common slips

- Setting the derivative of the squared distance with respect to t to zero and forgetting the root $t = 0$, or forgetting that $t^2 = p/a - 2$ has no solution when $p < 2a$: the case split is the whole point of part (ii).
- Giving $(mc - a)/(m\sqrt{m^2 + 1})$ when $mc < a$: this is negative, and the line actually cuts the parabola.
- Writing $d - b$ for the circle without checking its sign; when $b \geq d$ the curves meet and the answer is 0.

Beyond STEP

The nearest point in part (ii) is where CP is normal to the parabola, and the switch at $p = 2a$ is no accident: $(2a, 0)$ is the centre of curvature of the parabola at its vertex. Points beyond it have two nearest points off the axis, and the locus of such switching points, the evolute, is a classic object in the differential geometry of curves. Minimising a function over a restricted range, where the answer may sit at the boundary rather than at a stationary point, is the starting point of constrained optimisation. For more problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.